

2024

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer from **all** the Groups as directed.*

Group – A

(Objective Type Questions)

1. Choose the correct answer of the following :

1×5 = 5

(a) The dependency rules specified by data base designer are known as :

- (i) Functional dependencies
- (ii) Designer dependencies
- (iii) Databases
- (iv) None of these

- (b) Which of the following cannot be used to modify the data in Database ?
- (i) Delete
 - (ii) Update
 - (iii) Drop
 - (iv) Insert
- (c) In SQL, which command is used to change a table storage characteristics ?
- (i) Alter table
 - (ii) Modify table
 - (iii) Change table
 - (iv) All of these
- (d) _____ delete a data item from a database.
- (i) Insert
 - (ii) Drop
 - (iii) Delete
 - (iv) None of these
- (e) A relational database developer refer to record as :
- (i) Relation
 - (ii) Tuple
 - (iii) Data
 - (iv) None of these

2. Fill in the blanks with appropriate word : $1 \times 5 = 5$

- (a) A repository of information about a data base is known as _____.
- (b) _____ symbol is a terminator for SQL.
- (c) _____ command from SQL prompt is used to terminate from SQL.
- (d) Number of tuple in a relation is known as _____.
- (e) Composite key is a combination of _____ column.

Group – B

(Short-answer Type Questions)

3. Answer any four questions of the following :

$5 \times 4 = 20$

- (a) Write a PL / SQL block to find the Fibonacci series up to 25.
- (b) Write a PL / SQL block to find the factorial of a given number.
- (c) Write a PL / SQL code to accept your city name upper case and change into lower case.

- (d) Write a PL / SQL to find the prime number between 10 to 50.
- (e) Define functional and transitive dependency.
- (f) Explain the candidate key and super key with example.

Group – C

(Long-answer Type Questions)

4. Answer any four questions of the following :

$$10 \times 4 = 40$$

- (a) What is Normalization ? Explain the 1NF and 2NF with suitable example.
- (b) Explain the types of Triggers.
- (c) Explain the E-R Model with diagram.
- (d) Explain the DML, DDL and DCL with suitable example.
- (e) Explain the Arithmetic function of Oracle.
- (f) Write short notes on the following :
 - (i) Relational Model :
 - (ii) Data Dictionary
 - (iii) Revoke and Drop command

